

Amazon S3 Server-Side Encryption with AWS Key Management Service (SSE-KMS)

To Begin with the Lab

Summary of the Lab

In this lab, an S3 bucket was first prepared with two IAM users, Amit and Ravi, both having read-only access to the bucket through IAM policies. The bucket initially used the default **SSE-S3** encryption, and then a new **customer-managed KMS key** named Fox key was created in AWS KMS. The bucket was switched to **SSE-KMS** using that key, and the object inside the bucket was updated to inherit the bucket's encryption setting. Finally, the lab proved that Amit could still read and download the file because the KMS key was assigned to him, while Ravi could not decrypt or download the same file because he did not have permission to use the KMS key.

Understanding S3 Server-Side Encryption with AWS KMS

Server-side encryption with AWS KMS (SSE-KMS) protects S3 objects by encrypting them with a key managed in **AWS Key Management Service**. It exists so you can control who can decrypt the data, not just who can read the bucket. In this lab, the bucket was encrypted with a customer-managed KMS key, which meant the object could only be accessed by users who had both S3 read permission and KMS decrypt permission. The benefit is stronger security and better control, because AWS KMS lets you manage key usage, rotation, and audit activity through services like CloudTrail.

Example:

A bucket contains a file uploaded by the root user, and both Amit and Ravi have S3 read-only access. After the bucket is switched to SSE-KMS using the Fox key, Amit can still open and download the file because he is allowed to use the KMS key. Ravi cannot open or download the same file because S3 access alone is not enough—he also needs permission to decrypt with that KMS key.

Lab Steps

Step 1 – Create the prerequisite IAM users and S3 bucket

- Open **IAM**.
- Create two users: **Amit** and **Ravi**.
- Give both users console access and custom passwords.

IAM users (12) Info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID
cf-shashank1	/	0	1 hour ago	-	41 days	1 hour ago	[Redacted]
aws-aman	/	1	Yesterday	-	32 days	Yesterday	[Redacted]
aws-raman	/	0	6 days ago	-	25 days	6 days ago	[Redacted]
cf-nancy	/	0	21 days ago	-	21 days	21 days ago	-
syncuser1	/	0	15 days ago	-	15 days	15 days ago	-
cyberfrat-user1	/	1	4 days ago	-	4 days	4 days ago	[Redacted]
cyber-fratus2	/	1	-	-	4 days	-	-
cap1-praveen	/	1	20 hours ago	-	20 hours	20 hours ago	-
amit	/	0	-	-	11 minutes	-	-
ravi	/	0	-	-	6 minutes	-	-
iam_apicalluser1	/	0	11 days ago	-	-	-	[Redacted]
testuserapi1	/	0	5 days ago	-	-	-	[Redacted]

- Attach S3 read-only access to both users.

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

- ☐ Add user to group
- ☐ Copy permissions
- ☒ **Attach policies directly**

Permissions policies (1/2725)

Choose one or more policies to attach to your new user.

Filter by Type: All types (1 match)

Policy name	Type	Attached entities
☒ AmazonS3ReadOnlyAccess	AWS managed	37

► Set permissions boundary - optional

Cancel Previous **Next**

- Open Amazon S3.
- Create one bucket and keep the default encryption as SSE-S3 for now.

Amazon S3 > Buckets > Create bucket

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)

cloudfreeks-ssekms

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is controlled by bucket policies and IAM permissions.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects is controlled by bucket policies and IAM permissions.

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- Upload one test file into the bucket.

Amazon S3 > Buckets > cloudfreeks-ssekms

cloudfreeks-ssekms [Info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

[Find objects by prefix](#)

☐	Name	Type	Last modified	Size	Storage class
☐	sample_file.txt	txt	July 21, 2026, 13:10:46 (UTC+05:30)	29.0 B	Standard

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Step 2 – Verify S3 access before KMS encryption

- Log in as Amit in an incognito window.

ap-southeast-2.signin.aws.amazon.com/oauth?client_id=am%3Aaws%3Asignin%3A%3A%3Aconsole%2Fcanvas&code_challenge=mG...

Provide feedback Multi-session disabled English

aws

IAM user sign in

Account ID or alias (Don't have?)

☐ Remember this account

IAM username
amit

Password

☐ Show Password [Having trouble?](#)

Sign in

[Sign in using root user email](#)

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- Open the bucket.
- Confirm that Amit can read and download the object.

cloudfreeks-ssekms

Objects Metadata Properties Permissions Metrics Management File systems - new Access Points

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

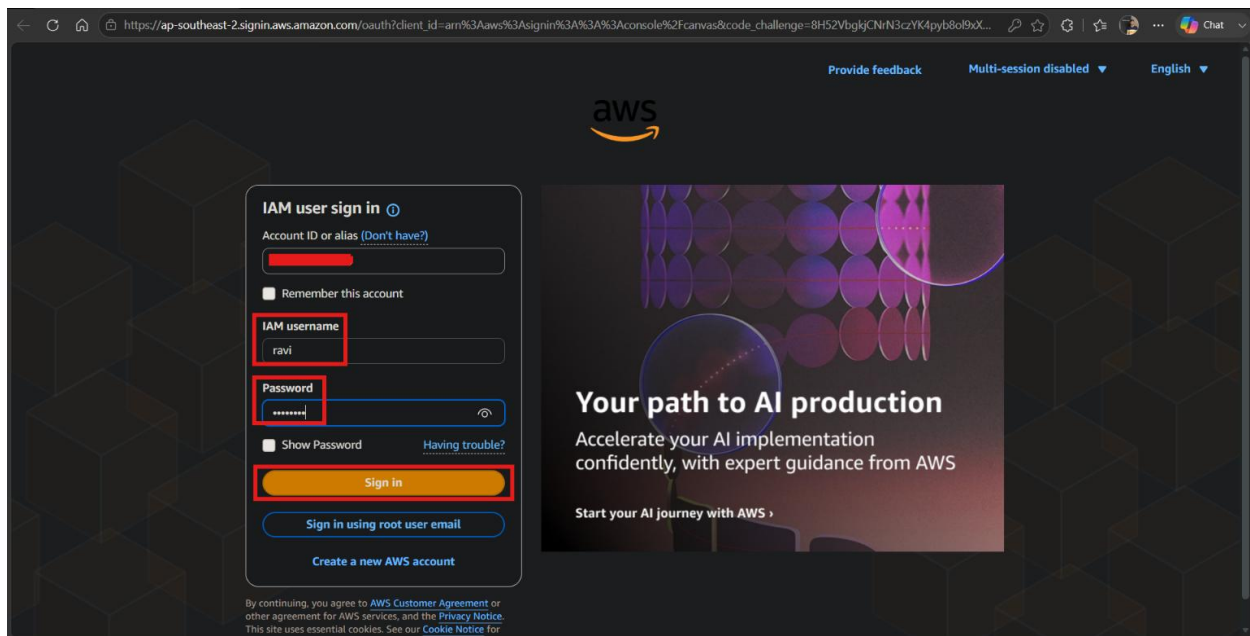
Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	sample_file.txt	txt	July 21, 2026, 13:10:46 (UTC+05:30)	29.0 B	Standard

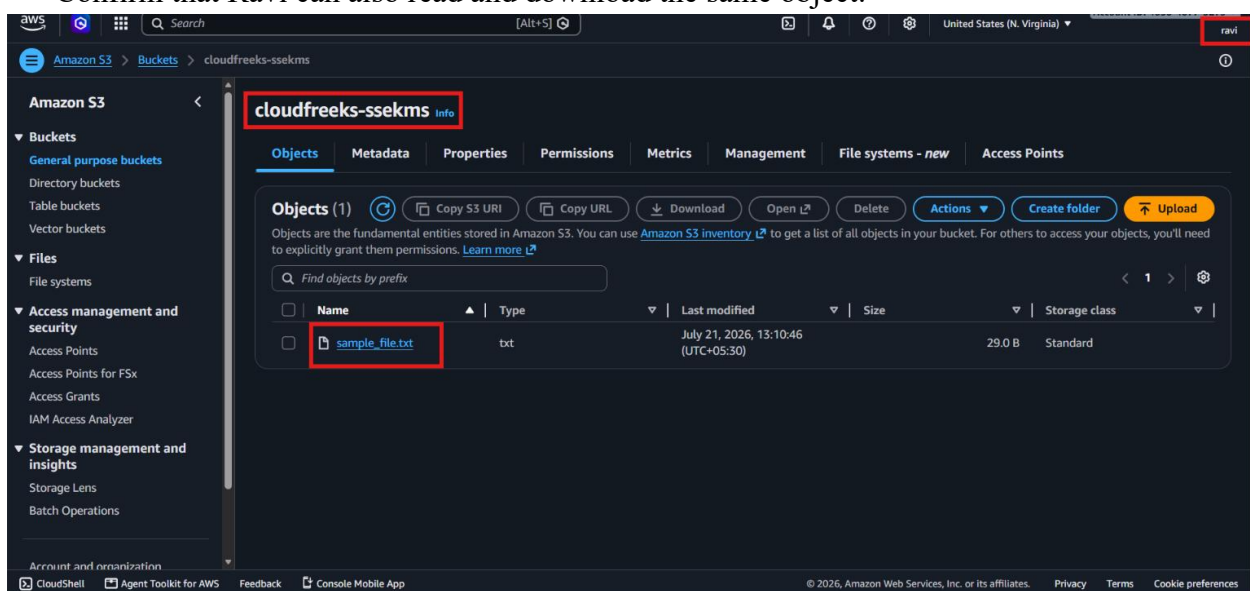
CloudShell Agent Toolkit for AWS Feedback Console Mobile App

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- Log in as **Ravi** in another private browser window.

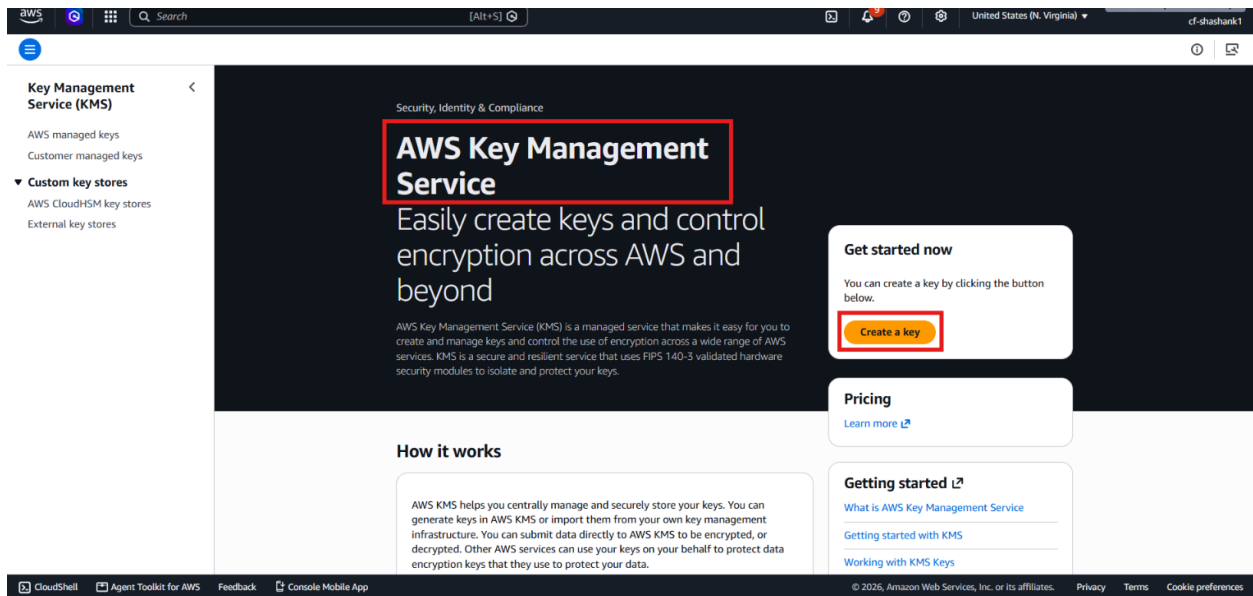


- Confirm that Ravi can also read and download the same object.

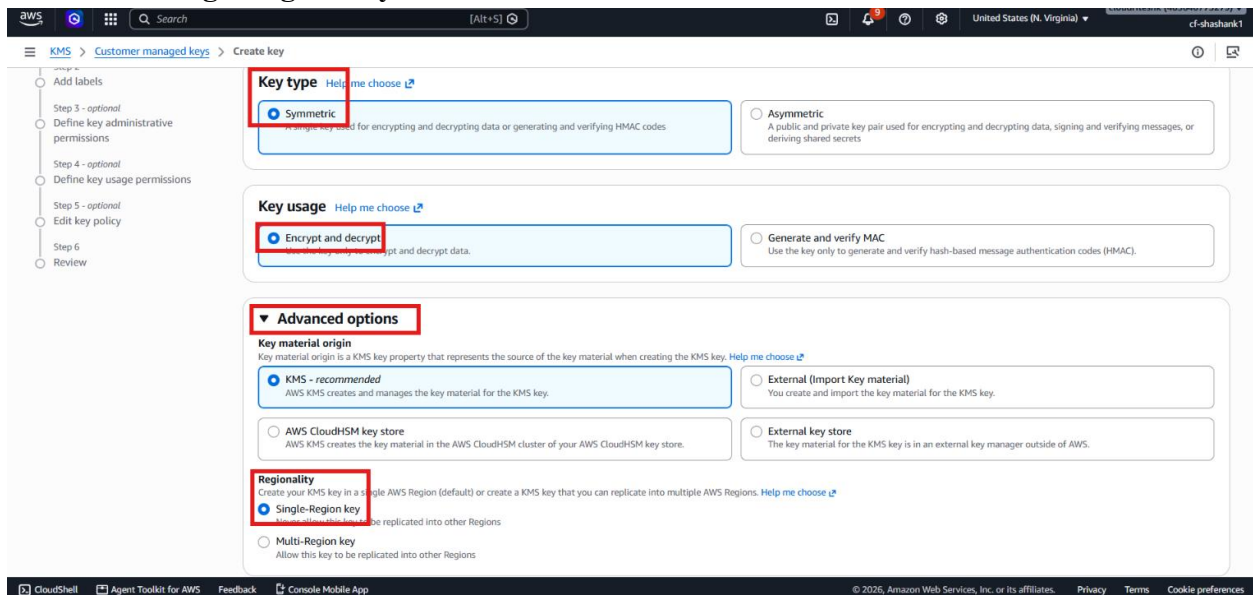


Step 3 – Create a customer-managed KMS key

- Log in as the root user.
- Open AWS KMS.
- Click **Create key**.



- Choose a **symmetric** key.
- Select a **single-region** key.



- Name the key: cloudfreeks-s3-key.
- Leave rest settings as default.

Step 2: Add labels

Step 3 - optional: Define key administrative permissions

Step 4 - optional: Define key usage permissions

Step 5 - optional: Edit key policy

Step 6: Review

Alias
You can change the alias at any time. [Learn more](#)

Alias:

Description - optional
You can change the description at any time.

Description:

Tags - optional
You can use tags to categorize and identify your KMS keys and help you track your AWS costs. When you add tags to AWS resources, AWS generates a cost allocation report for each tag. [Learn more](#)

This key has no tags.

[Add tag](#)

You can add up to 50 more tags.

[Cancel](#) [Skip to Review](#) [Previous](#) [Next](#)

- Assign the key to Amit only.

Introducing the new Create key experience
We've improved the create key experience with an enhanced policy editor. [Let us know what you think](#) or you can [use the old experience](#).

Step 1: Configure key

Step 2: Add labels

Step 3 - optional: Define key administrative permissions

Step 4 - optional: Define key usage permissions

Step 5 - optional: Edit key policy

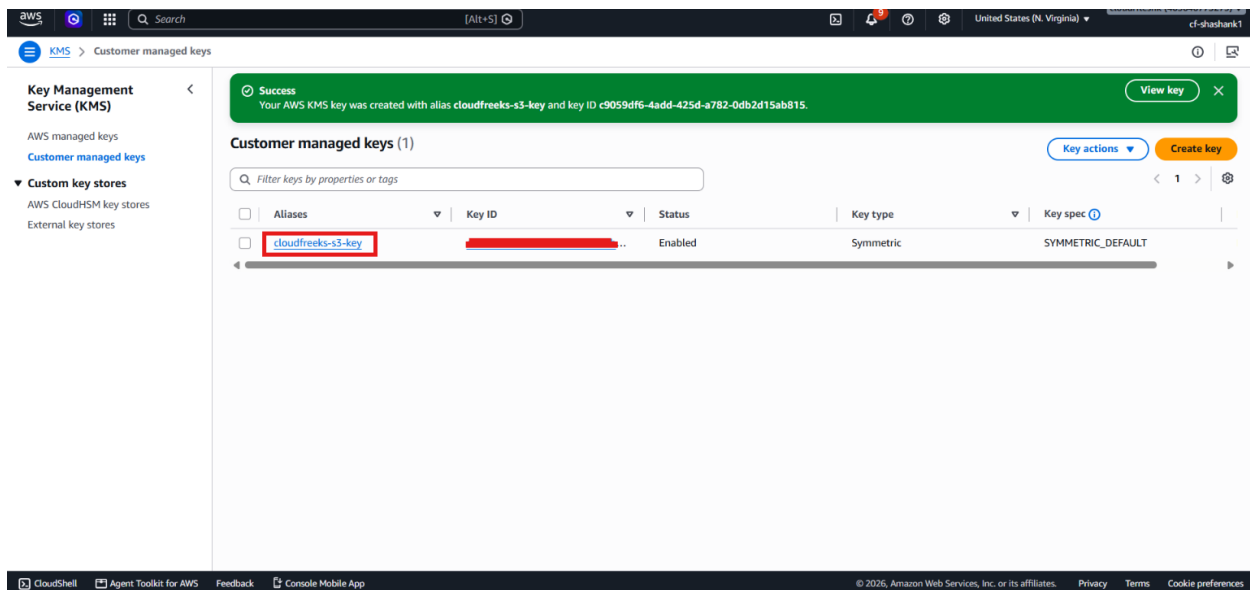
Step 6: Review

Define key administrative permissions - optional

Key administrators (1/517)
Select the IAM users and roles authorized to manage this key via the KMS API. These administrators will be added to the key policy under the statement identifier (Sid) 'Allow administration of the key'. Modifying this Sid might impact the console's ability to update the administrator statement in the key policy. [Learn more](#)

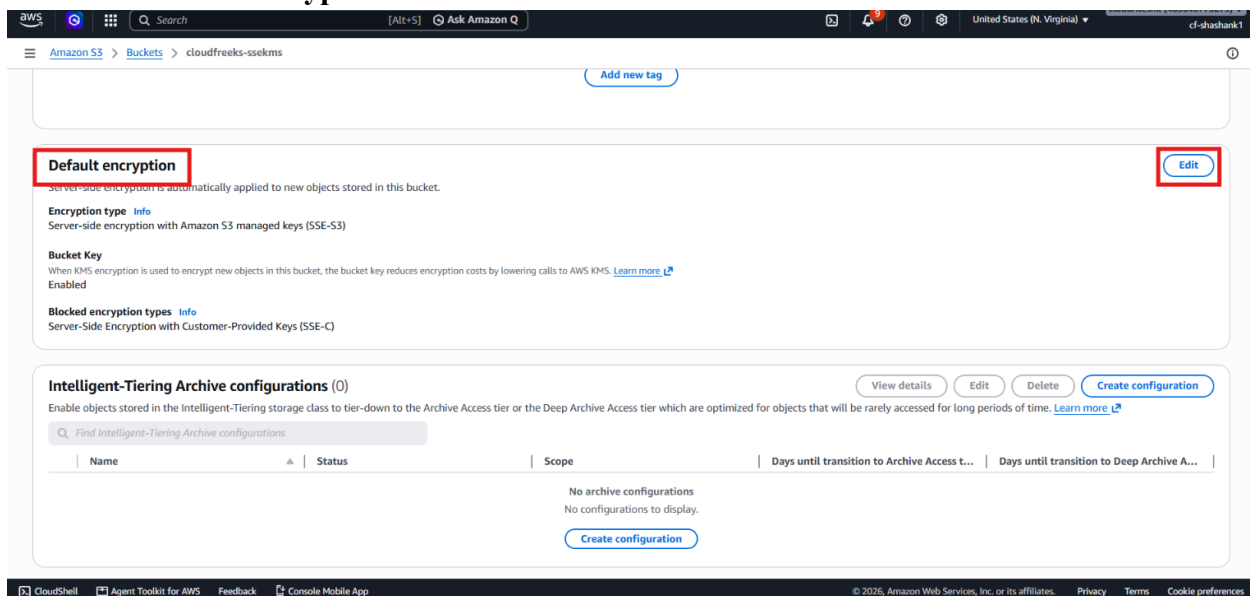
☐	Name	Path	Type
☒	amit	/	User
☐	aws-aman	/	User
☐	aws-raman	/	User
☐	cap1-praveen	/	User
☐	cf-nancy	/	User
☐	cf-shashank1	/	User
☐	cyber-fratusr2	/	User
☐	cyberfrat-user1	/	User
☐	iam_apicalluser1	/	User

- Finish creating the key.



Step 4 – Encrypt the S3 bucket with SSE-KMS

- Return to the S3 bucket.
- Open **Properties**.
- Go to **Default encryption**.



- Change the encryption type to **Server-side encryption with AWS KMS**.
- Select the customer-managed key Fox key.

Default encryption
Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

- ☐ Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ☒ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the [Storage tab of the Amazon S3 pricing page](#).

AWS KMS key [Info](#)

- ☒ Choose from your AWS KMS keys
- ☐ Enter AWS KMS key ARN

Available AWS KMS keys

aws/s3

cloudfreeks-s3-key

☐ Disable

☒ Enable

Blocked encryption types [Info](#)

- ☒ SSE-C (Server-Side Encryption with Customer-Provided Keys)

[Cancel](#) [Save changes](#)

- Save the changes.
- Update the object so it inherits the bucket's encryption setting by opening properties of object.

sample_file.txt

The following bucket properties and object management configurations impact the behavior of this object.

Bucket properties

Bucket Versioning
When enabled, multiple variants of an object can be stored in the bucket to easily recover from unintended user actions and application failures.

Disabled

Bucket "cloudfreeks-ssekms" doesn't have Bucket Versioning enabled
We recommend that you enable Bucket Versioning to help protect against unintentionally overwriting or deleting objects. [Learn more](#)

[Enable Bucket Versioning](#)

Management configurations

Replication status
When a replication rule is applied to an object the replication status indicates the progress of the operation.

-

[View replication rules](#)

Expiration rule
You can use a lifecycle configuration to define expiration rules to schedule the removal of this object after a pre-defined time period.

-

Expiration date
The object will be permanently deleted on this date.

-

Storage class
Amazon S3 offers a range of storage classes designed for different use cases. [Learn more](#) or see [Amazon S3 pricing](#)

Storage class
Standard

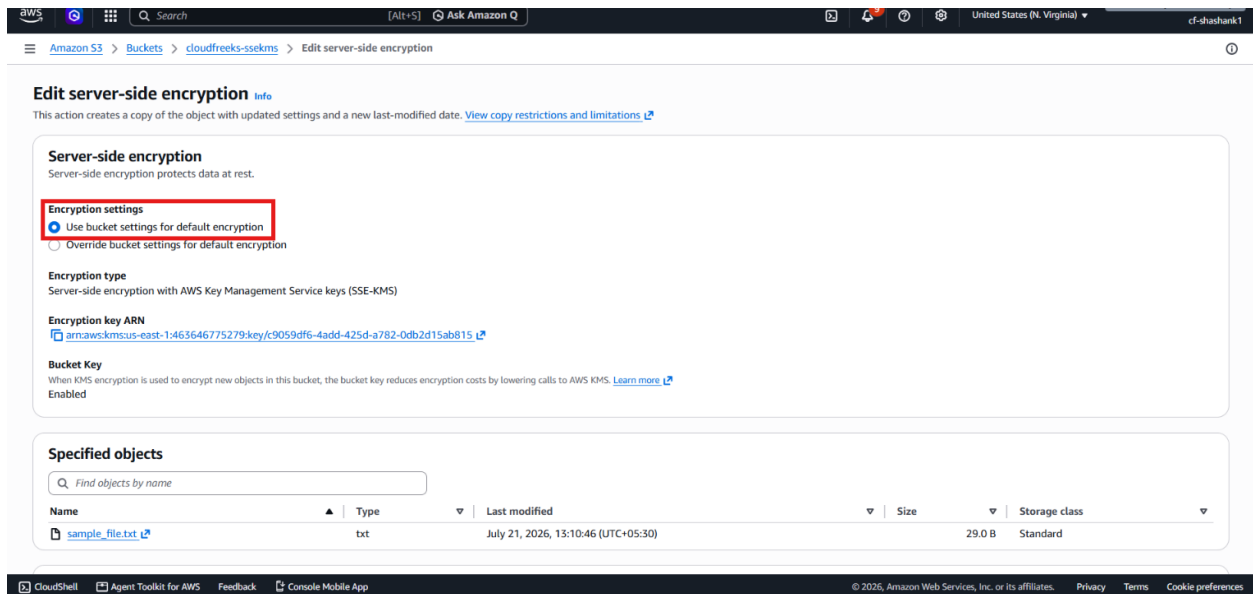
Server-side encryption settings [Info](#)

Server-side encryption protects data at rest.

Encryption type [Info](#)
Server-side encryption with Amazon S3 managed keys (SSE-S3)

[Edit](#)

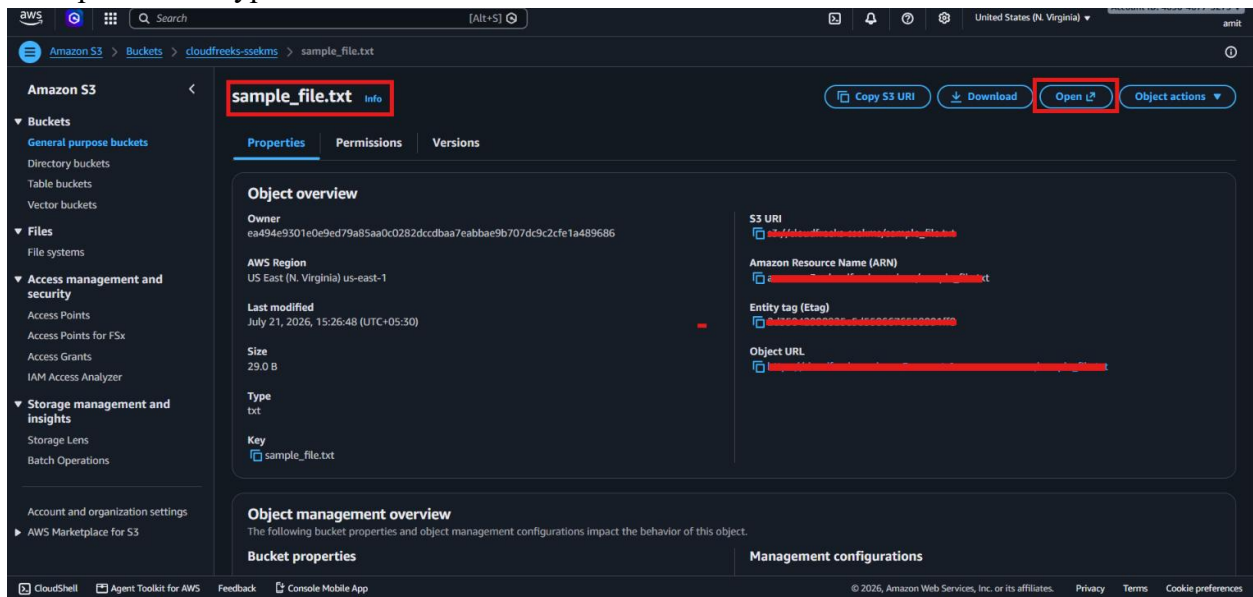
- Now change the encryption to Use bucket settings.



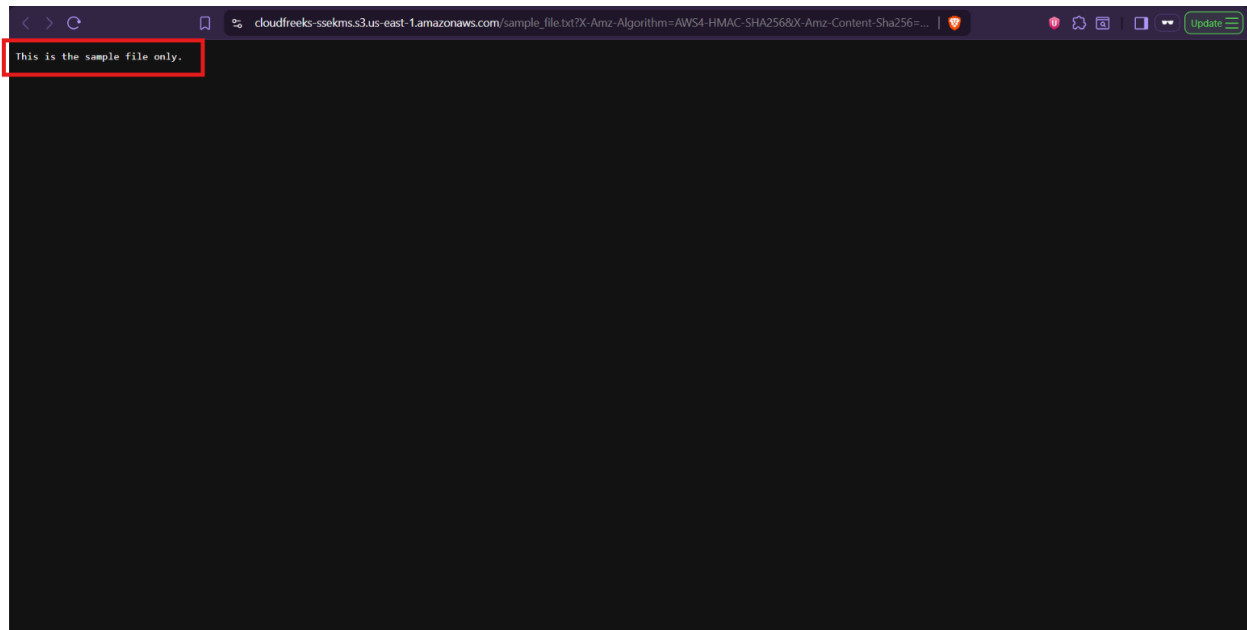
- Save changes.

Step 5 – Test access with Amit

- Log in as Amit.
- Open the encrypted bucket.



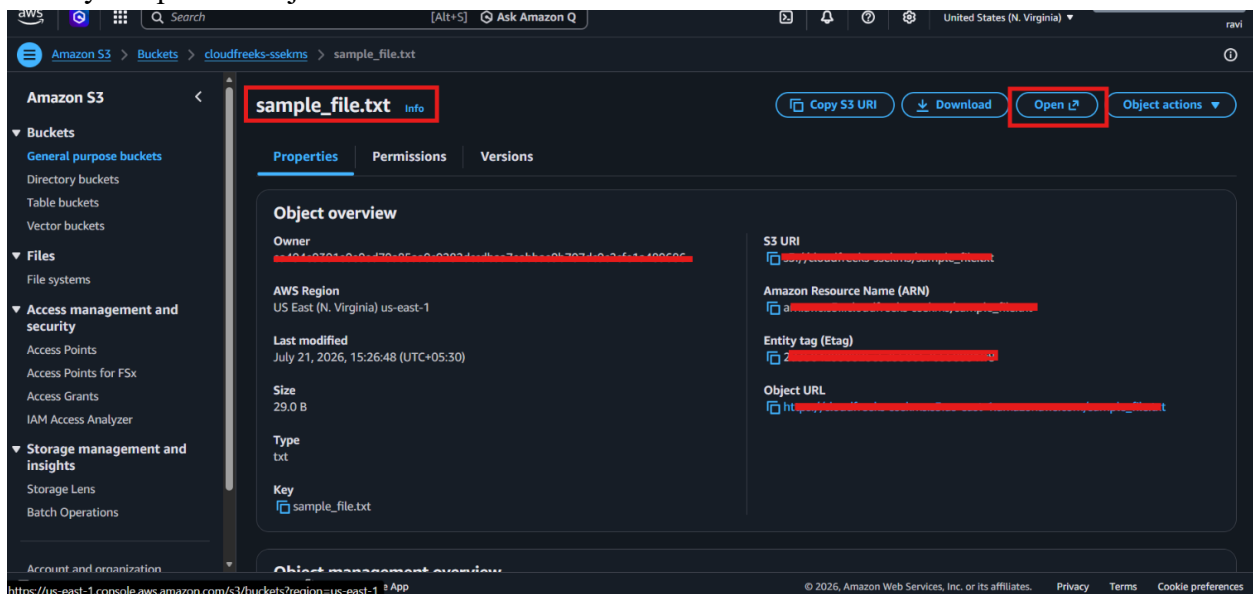
- Confirm that Amit can still open the object.



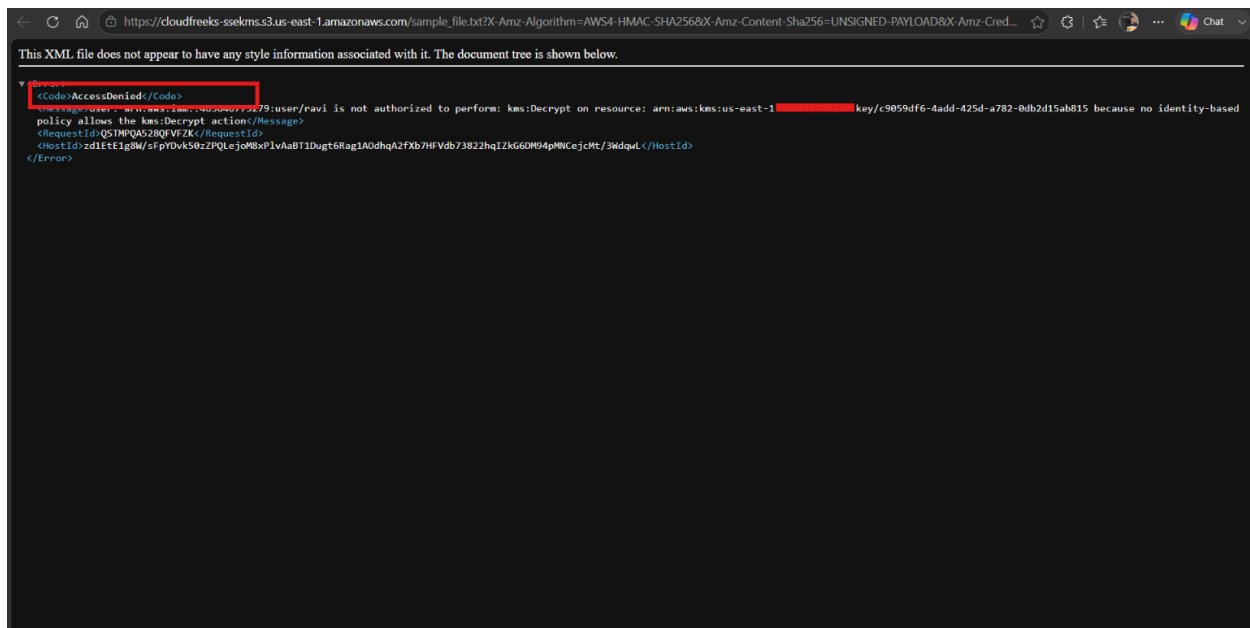
- Confirm that Amit can also download the object successfully.
- Observe that decryption happens transparently for Amit.

Step 6 – Test access with Ravi

- Log in as **Ravi**.
- Open the same encrypted bucket.
- Try to open the object.



- Confirm that access fails because Ravi does not have permission to use the KMS key.



- Try to download the object and confirm that decrypt access is denied.

Verification

- Verified that both Amit and Ravi could access the S3 bucket before KMS encryption.
- Verified that the bucket default encryption was changed from **SSE-S3** to **SSE-KMS**.
- Verified that a customer-managed KMS key named Fox key was created successfully.
- Verified that Amit could still decrypt and access the file because he had KMS permission.
- Verified that Ravi could not open or download the file because he lacked KMS decrypt permission.
- Verified that SSE-KMS provides stronger access control and auditability than default S3 encryption.